

ANSWER SCHEME

SECTION A

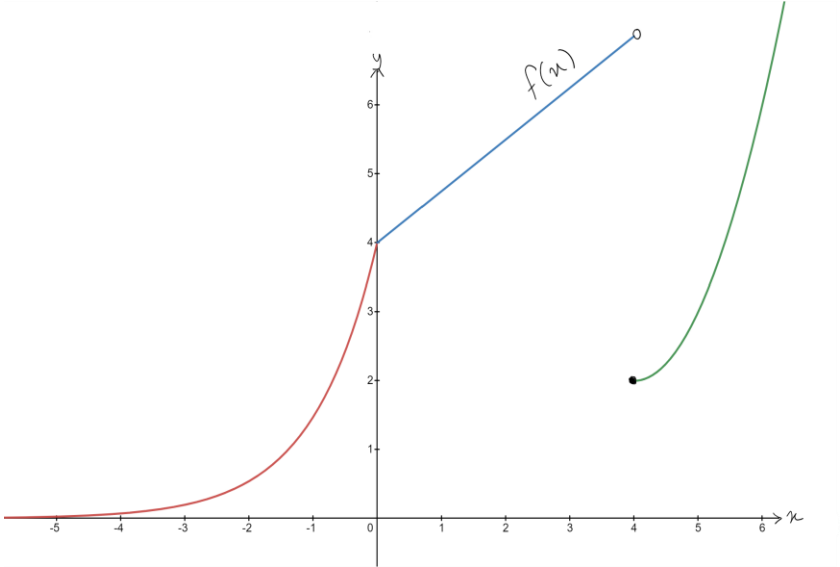
Question	Working Schemes	Remarks
1.	a) $\lim_{x \rightarrow 1} \frac{5x^2 - px}{2x^2 - x} = 1$ $\lim_{x \rightarrow 1} \frac{x(5x - p)}{x(2x - 1)} = 1$ $\frac{5(1) - p}{2(1) - 1} = 1$ $p = 4$ b) $\lim_{x \rightarrow +\infty} \left(\frac{8x^2 + 8x - 2}{x^2 + 1} \right)^{\frac{1}{3}}$ $= \left(\lim_{x \rightarrow +\infty} \frac{8x^2 + 8x - 2}{x^2 + 1} \right)^{\frac{1}{3}}$ $= \left(\lim_{x \rightarrow +\infty} \frac{8x^2/x^2 + 8x/x^2 - 2/x^2}{x^2/x^2 + 1/x^2} \right)^{\frac{1}{3}}$ $= \left(\lim_{x \rightarrow +\infty} \frac{8 + 8/x - 2/x^2}{1 + 1/x^2} \right)^{\frac{1}{3}}$ $= \left(\frac{8 + 0 - 0}{1 + 0} \right)^{\frac{1}{3}}$ $= 2$ c) $\lim_{x \rightarrow 2^-} \frac{(3x + 2) 2 - x }{2 - x}$ $= \lim_{x \rightarrow 2^-} \frac{(3x + 2)(2 - x)}{2 - x}$ $= 3(2) + 2$ $= 8$	K1 – factorise and simplify K1- subs $x = 1$ J1 K1– divide each term with x^2 K1 –sub J1 K1 – sub correct function and simplify K1-sub J1
		9 marks

2.	a) $u' = 6x \quad v' = \frac{24x^2}{(2x^3 - 5)}$ $\frac{dy}{dx} = 6x \cdot \ln(2x^3 - 5)^4 + \frac{24x^2}{(2x^3 - 5)} \cdot 3x^2$ $\frac{dy}{dx} = 6x \cdot \ln(2x^3 - 5)^4 + \frac{72x^2}{(2x^3 - 5)}$ b) $u' = -\sin x \quad v' = \frac{3}{2}(1 + \sin 3x)^{-\frac{1}{2}} \cos 3x$ $\frac{dy}{dx} = \frac{-\sin x(1 + \sin 3x)^{\frac{1}{2}} - \frac{3}{2}(1 + \sin 3x)^{-\frac{1}{2}} \cos 3x(1 + \cos x)}{(1 + \sin 3x)}$ $\frac{dy}{dx} = \frac{-2 \sin x(1 + \sin 3x) - 3 \cos 3x(1 + \cos x)}{2(1 + \sin 3x)^{\frac{3}{2}}}$	K1 – u' and v' K1 - product K1 J1 K1 – u' and v' K1 – quotient K1 – single fraction K1 – simplify J1
		9 marks
3.	$z^2 = x^2 + y^2$ $2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$ $z \frac{dz}{dt} = x \frac{dx}{dt} + y \frac{dy}{dt}$ at $x = 6, y = 8$ $z^2 = 6^2 + 8^2$ $z = 10$ $(10) \frac{dz}{dt} = (6)(16) + 8(13)$ $\frac{dz}{dt} = 20 \text{ km / h}$	K1 K1 K1 K1 J1 K1 J1
		7 marks

SECTION B

Question	Working Schemes	Remarks
1.	$ z = a + 4\sqrt{2} + (a - 4\sqrt{2})i = 2\sqrt{21}$ $\sqrt{(a + 4\sqrt{2})^2 + (a - 4\sqrt{2})^2} = 2\sqrt{21}$ $a^2 + 8a\sqrt{2} + 32 + a^2 - 8a\sqrt{2} + 32 = 84$ $2a^2 = 20$ $a = \pm\sqrt{10}$ Since $a > 0 \therefore a = \sqrt{10}$ $\sqrt{10} + 4\sqrt{2} + (\sqrt{10} - 4\sqrt{2})i$ $\arg z = \tan^{-1}\left(\frac{\sqrt{10} - 4\sqrt{2}}{\sqrt{10} + 4\sqrt{2}}\right)$ $= -0.276$ $\therefore \sqrt{10} + 4\sqrt{2} + (\sqrt{10} - 4\sqrt{2})i = 2\sqrt{21}(\cos(-0.276) + i \sin(-0.276))$	K1 – sub a and b into modulus formula K1- try to find a J1 K1✓-try to find arg z J1
		5 marks
2.	a) $1 + \sqrt{x+3} = \sqrt{2x-1}$ $(1 + \sqrt{x+3})^2 = (\sqrt{2x-1})^2$ $1 + 2\sqrt{x+3} + x + 3 = 2x - 1$ $2\sqrt{x+3} = x - 5$ $4(x+3) = x^2 - 10x + 25$ $4x + 12 = x^2 - 10x + 25$ $x^2 - 14x + 13 = 0$ $(x-13)(x-1) = 0$ $x = 13, x = 1$ <i>check, x=13</i>	K1 – SBS K1 – collect surd terms K1 – SBS K1 – solve quadratic J1

	b) $\left \frac{x+2}{x-4} \right < 5$ $\frac{x+2}{x-4} > -5 \quad \text{and} \quad \frac{x+2}{x-4} < 5$ $\frac{x+2}{x-4} + 5 > 0 \quad \frac{x+2}{x-4} + 5 < 0$ $\frac{6x-18}{x-4} > 0 \quad \frac{-4x+22}{x-4} < 0$ $6x-18 > 0, x-4 > 0 \quad -4x+22 > 0, x-4 > 0$ $x > 3, x > 4 \quad x < \frac{11}{2}, x > 4$ $x < 3 \cup x > 4 \quad x < 4, x > \frac{11}{2}$ $\therefore \{x : x < 3 \text{ or } x > \frac{11}{2}\}$	K1 – definition K1- collect factors (either 1) K1 – method 1st part K1 – method 2nd part K1 – combine J1
		11 marks
3.	a) $f(x) = 2 - \ln 3-x $ $D_f = (-\infty, 3) \quad R_f = \mathbb{R}$ b) $f(x_1) = f(x_2)$ $2 - \ln 3-x_1 = 2 - \ln 3-x_2 $ $\ln 3-x_1 = \ln 3-x_2 $ $3-x_1 = 3-x_2$ $x_1 = x_2$ Since $f(x)$ is one to one $\therefore f^{-1}(x)$ is exist. $f(x) = 2 - \ln 3-x $ $ff^{-1}(x) = x$ $2 - \ln 3-f^{-1}(x) = x$ $f^{-1}(x) = 3 - e^{2-x}$	B1 B1 K1 – use algebraic/graph J1 K1 J1
		6 marks

4.	$2f(x)+5=\frac{x+2}{3}$	K1
	$2f(x)=\frac{x-13}{3}$	K1
	a) $f(x)=\frac{x-13}{6}$	J1
	$f(x-1)=\frac{x-1-13}{6}$	K1
	$=\frac{x-14}{6}$	J1
	b)	
	i.	
		R1 R1 R1 R1 R1
	ii.	
	$D_h = R_{h^{-1}} = (-\infty, \infty)$	B1B1
	$R_h = D_{h^{-1}} = (0, \infty)$	B1B1
		14 marks

5.	a) $f(x) = \frac{x+4}{\sqrt{x^2-25}}$ $x^2 - 25 = 0$ $\therefore x = 5, -5$ $\lim_{x \rightarrow 5^+} \frac{5+4}{\sqrt{5^2-25}} = \frac{9}{0^+} = +\infty$ $\lim_{x \rightarrow 5^-} \frac{5+4}{\sqrt{5^2-25}} \text{ does not exist}$ $\lim_{x \rightarrow -5^+} \frac{-5+4}{\sqrt{5^2-25}} \text{ does not exist}$ $\lim_{x \rightarrow -5^-} \frac{-5+4}{\sqrt{5^2-25}} = \frac{-1}{0^+} = -\infty$ $\therefore$ the vertical asymptotes are $x = 5$ and $x = -5$ $\lim_{x \rightarrow +\infty} \frac{\frac{x}{x} + \frac{4}{x}}{\sqrt{\frac{x^2}{x^2} - \frac{25}{x^2}}} = \frac{1}{1} = 1$ $\lim_{x \rightarrow -\infty} \frac{\frac{x}{-x} + \frac{4}{-x}}{\sqrt{\frac{x^2}{x^2} - \frac{25}{x^2}}} = \frac{-1}{1} = -1$ $\therefore$ the horizontal asymptotes are $y = 1$ and $y = -1$ b) $h(x) = \begin{cases} e^x + 2, & x < 0 \\ x^2 + 3, & x \geq 0 \end{cases}$ 1. $h(x) = x^2 + 3$ $h(0) = 0^2 + 3 = 3$ 2. $\lim_{x \rightarrow 0^-} e^x + 2 = e^0 + 2 = 3$ $\lim_{x \rightarrow 0^+} x^2 + 3 = 0 + 3 = 3$ $\therefore \lim_{x \rightarrow 0} f(x) = 3$ 3. since $h(0) = \lim_{x \rightarrow 0} f(x) = 3$ $\therefore f(x)$ is continuous at $x = 0$	K1 K1 – find either one lim $x \rightarrow 5$ K1 – both correct K1 - find either one lim $x \rightarrow -5$ K1 – both correct J1 K1 - find lim $x \rightarrow +\infty$ K1 - find lim $x \rightarrow -\infty$ J1 K1-find f(x) K1- find lim J1 K1 J1
		14 marks

6.	i) $y = \frac{x}{2x+1}$ $x = 2xy + y$ $\frac{dx}{dy} = 2y \frac{dx}{dy} + 2x + 1$ $\frac{dx}{dy} - 2y \frac{dx}{dy} = 2x + 1$ $\frac{dx}{dy} - 2y \frac{dx}{dy} = 2x + 1$ $\frac{dx}{dy} = \frac{2x+1}{1-2y}$ ii) $\frac{d^2x}{dy^2} = \frac{(2 \frac{dx}{dy})(1-2y) - [-2(2x+1)]}{(1-2y)^2}$ $= \frac{2(\frac{2x+1}{1-2y})(1-2y) - [-2(2x+y)]}{(1-2y)^2}$ $= \frac{2(1-2y)(2x+1)}{(1-2y)^3} + \frac{4x+2}{(1-2y)^2}$ $= \frac{8x-8y-16xy+4}{(1-2y)^3}$	K1-LHS K1 – RHS K1 – try to get dx/dy J1 K1 – quotient K1 – subs dx/dy K1 – single fraction J1
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	b) $\frac{dx}{dt} = \frac{2t}{(2-t^2)^{\frac{3}{2}}}$ $\frac{dy}{dt} = \frac{-2}{t^2(2-t^2)^{\frac{1}{2}}}$ $\frac{dy}{dx} = \frac{-(2-t^2)}{t^3}$ $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left[\frac{-(2-t^2)}{t^3} \right]}{\frac{2t}{(2-t^2)^{\frac{3}{2}}}}$ $= \frac{(6-t^2)(2-t^2)^{\frac{3}{2}}}{2t^5}$ when $t=1$ $\frac{dy}{dx} = \frac{-(2-1)}{1} = -2$ when $t=1$, $\frac{d^2y}{dx^2} = \frac{(6-1)(1)}{2} = \frac{5}{2}$	K1 K1 K1 – by chain rule K1 K1 K1 subs 1 into either J1 J1
	16 marks	

7.	$4y + 2\pi x = 8$ a) $y = \frac{8 - 2\pi x}{4}$ $= 2 - \frac{\pi x}{2}$ $A = \pi x^2 + y^2$ $= \pi x^2 + \left(2 - \frac{\pi x}{2}\right)^2$ b) $= \pi x^2 + 4 - 2\pi x + \frac{\pi^2 x^2}{4}$ $= \left(\frac{\pi^2 + 4\pi}{4}\right)x^2 - 2\pi x + 4$ c) $\frac{dA}{dy} = \left(\frac{\pi^2 + 4\pi}{2}\right)x - 2\pi$ $\frac{dA}{dy} = 0$ $\left(\frac{\pi^2 + 4\pi}{2}\right)x - 2\pi = 0$ $\left(\frac{\pi^2 + 4\pi}{2}\right)x = 2\pi$ $x = 2\pi \times \frac{2}{\pi^2 + 4\pi}$ $x = \frac{4}{\pi + 4}$ $P_{circle} = 2\pi \left(\frac{4}{\pi + 4}\right)$ $= \frac{8\pi}{\pi + 4}$ $\% \text{wire used for circle} = \frac{\left(\frac{8\pi}{\pi + 4}\right)}{8} \times 100\%$ $= 43.99\%$	K1 J1 K1 K1 J1 K1 J1 K1 J1
		9 marks